



# ELVA Visual AI

Transforming Vision into Intelligence

**DETECT • VERIFY • OPTIMIZE • SECURE**



# What is ELVA?

## How It Works

- Ingest: 4K/AI IoT cameras, microscopes, and production line cameras.
- Understand: AI models for OCR, detection, anomaly spotting and classification.
- Act: Real-time alerts, dashboards, audit trails and automation workflows.
- Integrate: ERP, POS, RFID, barcode systems and access control.

## Key Differentiators

- Jewellery-specific AI datasets and pre-trained models.
- Micron-level defect and inclusion detection.
- Enterprise-grade audit and security controls.
- Hybrid deployment with edge + cloud architecture.

# Industry Challenges We Solve

## Jewellery Retail

- Shrinkage and pilferage detection.
- Counter crowd monitoring and staff analytics.
- Stock mismatch and display inconsistencies.
- Customer service and operational visibility.

## Manufacturing Plants

- Micro-defect and quality inspection.
- Packing and dispatch verification.
- Safety and PPE compliance monitoring.
- Traceability and production analytics.

AI-Driven Security • Operational Visibility • Process Automation • Compliance Intelligence

# Visual AI Use Cases Retail Industry - Challenges

## Store Intelligence

- Customer footfall analytics
- Counter-wise crowd analysis
- Salesperson activity tracking
- Suspicious behaviour detection
- Queue and wait-time monitoring

## Business Impact

- Reduced operational losses
- Improved customer engagement
- Optimized staff productivity
- Enhanced compliance visibility

AI-Driven Security • Operational Visibility • Process Automation • Compliance Intelligence

# Visual AI Use Cases Manufacturing Industry - Challenges

## Factory Intelligence

- AI-based PDI inspection
- OCR & barcode verification
- Surface defect detection
- Packing line validation
- Worker safety compliance

## Operational Benefits

- Reduced rework and wastage
- Higher quality consistency
- Automated inspection workflows
- Improved production visibility

AI-Driven Security • Operational Visibility • Process Automation • Compliance Intelligence

# Advanced AI Capabilities

## Core Technologies

- Computer Vision
- Object Detection
- OCR & Text Analytics
- Behavioural AI
- Predictive Intelligence

## Enterprise Integration

- ERP & BI integration
- Centralized AI dashboards
- Cloud & edge deployment
- Multi-location monitoring

AI-Driven Security • Operational Visibility • Process Automation • Compliance Intelligence

# Successfully Delivered Projects

## Core Use Cases

- AI-based retail intelligence platform.
- People counting & behavior analytics.
- Intrusion detection systems.
- Smart surveillance with automated alerts.
- Planogram compliance analytics.
- Workplace safety monitoring.
- PDI Monitoring.

## Clients

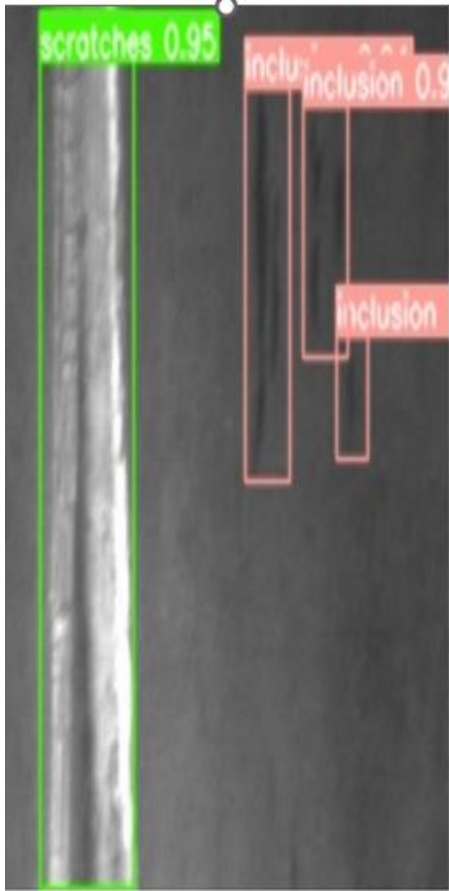
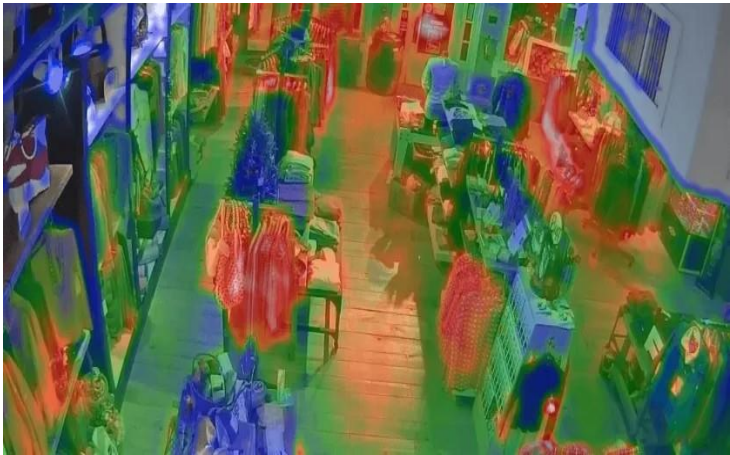
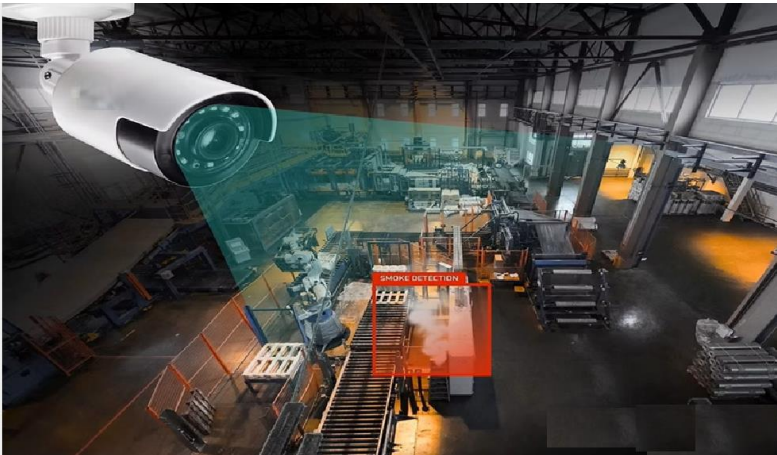
Senco Gold Limited.

Dhunseri Polyfilms Pvt. Limited.

Gibraltar Airsprings.



# Common Use Cases Detections





NETWORK OVERVIEW

All stores · Real-time operational status

● Store open

CUSTOMERS TODAY

37

Footfall IN across all stores

CURRENTLY INSIDE

17

Footfall IN minus OUT

STORES OPERATIONAL

1

1 configured

DEVICES ONLINE

8

8 commissioned

CROWD INCIDENTS

5

Only crowd volume 6 or higher

INTRUDER ALERTS

2

Closed-store security watch

STORE NETWORK

Select a store to begin monitoring operations

1 store

Senco Gold C006

Moulali, Kolkata

# Senco Gold C006

Stores → C006 · Moulali, Kolkata

V3 operations

Live operations

Customer Intelligence

Live Operations

Security & Alerts

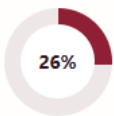
CUSTOMER INTELLIGENCE

## Product engagement and popularity

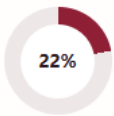
Routine interaction events become charts and summaries.  
They are not operational alerts.

### SECTION ENGAGEMENT

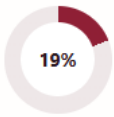
Interaction share



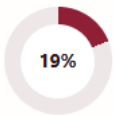
**Wedding\_section**  
7 interactions



**Diamond\_section**  
6 interactions



**Chain**  
5 interactions



**Everlite\_section**  
5 interactions

### ITEM POPULARITY MAP

Click a cell for details





# Senco Gold C006

Stores → C006 · Moulali, Kolkata

V3 operations

Live operations

Customer Intelligence

Live Operations

Security & Alerts

LIVE OPERATIONS

DATE

14 Aug 2026

● Refreshes automatically

## Real-time store intelligence

Current activity, evidence and operational exceptions are monitored inside this store dashboard. Routine records are grouped so repeated detections do not overwhelm the operator.

CURRENTLY INSIDE

17

Backend-confirmed occupancy

FOOTFALL TODAY

37

20 exited

PRODUCT INTERACTIONS

27

Confirmed analytics records

CROWD ALERTS

5

Triggered only at 6+ customers

INTRUDER INCIDENTS

2

Critical security events

SECTION INTELLIGENCE

Select a section to filter the evidence stream

Bangle\_section

CROWDED

4

Chain

CROWDED

7

Wedding\_section

CROWDED

7

# Senco Gold C006

Stores — C006 · Moulali, Kolkata

V3 operations

Live operations

Customer Intelligence

Live Operations

Security & Alerts

SECURITY & ALERTS

## Only actionable live incidents

Routine events remain analytics. Only crowd volume above 5 and intruder incidents appear here.

CROWD ALERTS

5

Confirmed at 6+ customers

INTRUDER ALERTS

2

Closed-store person detection

HIGHEST CROWD VOLUME

9

Diamond\_section

LIVE INCIDENT TIMELINE

Crowd and intruder only

8

CROWD ALERT **Wedding\_section — 8 customers**  
14 Aug · 02:28 pm · Single incident

View evidence

9

CROWD ALERT **Diamond\_section — 9 customers**  
14 Aug · 01:52 pm · Single incident

View evidence

7

CROWD ALERT **Everlite\_section — 7 customers**  
14 Aug · 01:50 pm · Single incident

View evidence

Wedding Anchor24 present=24/24 empty=0 unknown=0 obs=1.00 quality=OK



# SENCO GOLD C006 MOULALI – DISTRIBUTED AI EDGE ARCHITECTURE

Raspberry Pi Cameras with Local AI Processing • Events & Insights to Central Dashboard

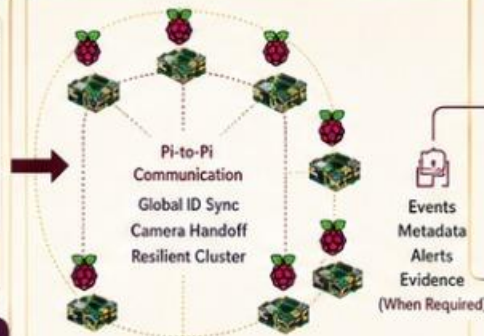
## 1. STORE CAMERAS (PI CAMERAS)



LOCAL AI ON EACH RASPBERRY PI

- YOLO Detection
- Tracking
- Staff Classification
- Journey Tracking
- Footfall Counting
- Local Cache
- Event Generation
- Pi-to-Pi Communication

## 2. DISTRIBUTED PI CLUSTER (EQUAL NODES)



## 3. CENTRAL DASHBOARD SERVER



## 4. STORE INTELLIGENCE - LIVE OVERVIEW



Crowd Level: Normal (0-4), Busy (5-7), Crowded (>7) | Footfall (Today): IN: 248, OUT: 231, Net: +17

## 5. KEY METRICS (TODAY)

|                |                |                 |                          |
|----------------|----------------|-----------------|--------------------------|
| Total Footfall | 248 IN         | Total Footfall  | 231 OUT                  |
| Net Footfall   | +17 (IN - OUT) | Net Engagement  | 02:41:18 (Today)         |
| Avg Dwell Time | 04:32 mins     | Engagement Rate | 68% (engaged vs visited) |

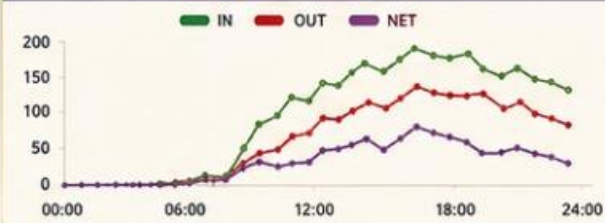
## 7. ZONE WISE CROWD (LIVE)

|          |   |         |
|----------|---|---------|
| WEDDING  | 7 | CROWDED |
| BANGLE 1 | 6 | BUSY    |
| DIAMOND  | 5 | BUSY    |
| BANGLE 2 | 4 | NORMAL  |
| CHAIN    | 3 | NORMAL  |
| EVERLITE | 2 | NORMAL  |

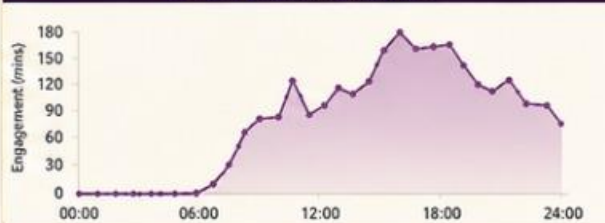
## 9. TOP ENGAGED SECTIONS (TODAY)

|                  |          |
|------------------|----------|
| Wedding Section  | 02:15:34 |
| Bangle Section   | 01:48:22 |
| Diamond Section  | 01:22:18 |
| Chain Section    | 00:58:44 |
| Everlite Section | 00:35:12 |

## 6. FOOTFALL TREND



## 8. NET ENGAGEMENT TREND



## 10. LIVE ALERTS

|                 |                               |          |
|-----------------|-------------------------------|----------|
| Crowd Alert     | Crowd: 7                      | 11:42:10 |
| Wedding Section |                               |          |
| Intruder Alert  | After Hours Movement Detected | 12:15:45 |
| Staff On Mobile | Bangle 1                      | 12:05:22 |

## 11. SMART SUMMARIES (TODAY)

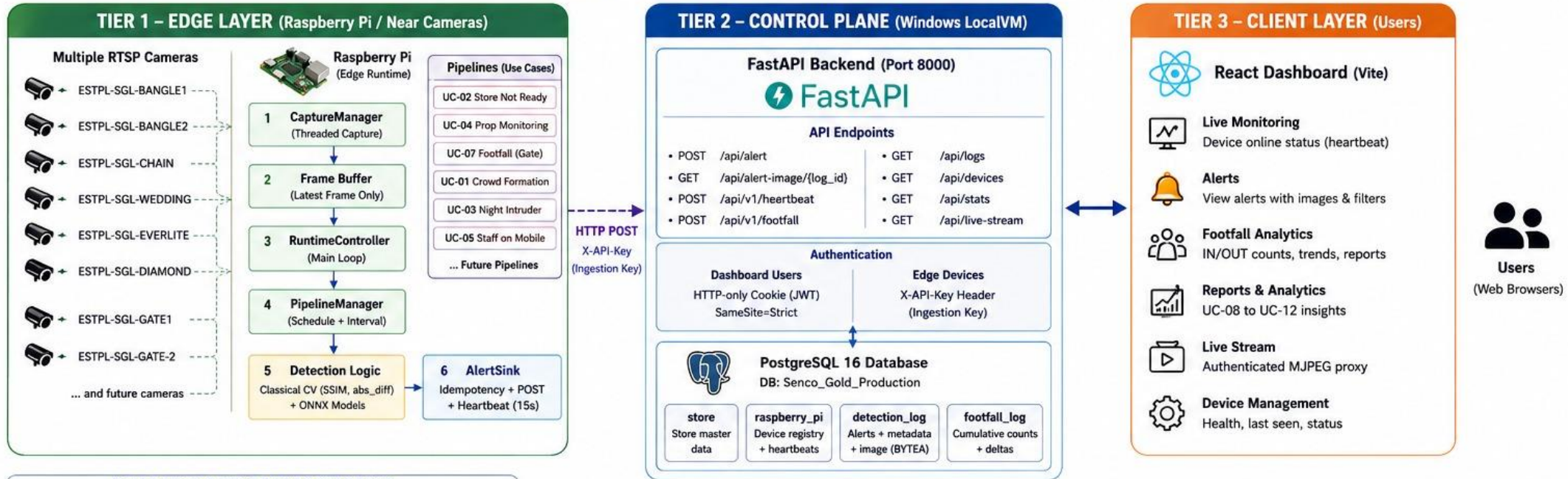
|                                                         |                                                     |                                                   |                                              |                                                  |                                                        |
|---------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------------------|
| Peak Footfall<br>12:00 PM ~ 2:00 PM<br>IN: 96   OUT: 84 | Busiest Section<br>Wedding Section<br>Peak Crowd: 7 | Highest Engagement<br>Wedding Section<br>02:15:34 | Conversion Insight<br>Engagement Rate<br>68% | Staff Insight<br>Active Staff: 18<br>On Desk: 14 | Security Insight<br>No Intrusion Till Now<br>All Clear |
|---------------------------------------------------------|-----------------------------------------------------|---------------------------------------------------|----------------------------------------------|--------------------------------------------------|--------------------------------------------------------|

## 12. SYSTEM STATUS

|                              |                         |                       |                          |
|------------------------------|-------------------------|-----------------------|--------------------------|
| All Pi Nodes Online<br>7 / 7 | Cluster Sync<br>Healthy | Last Sync<br>12:14:05 | Server Status<br>Healthy |
|------------------------------|-------------------------|-----------------------|--------------------------|

# SENCO GOLD & DIAMONDS – AI SURVEILLANCE SYSTEM (UPDATED DESIGN)

Scalable • Secure • Modular • Real-time • Multi-Camera

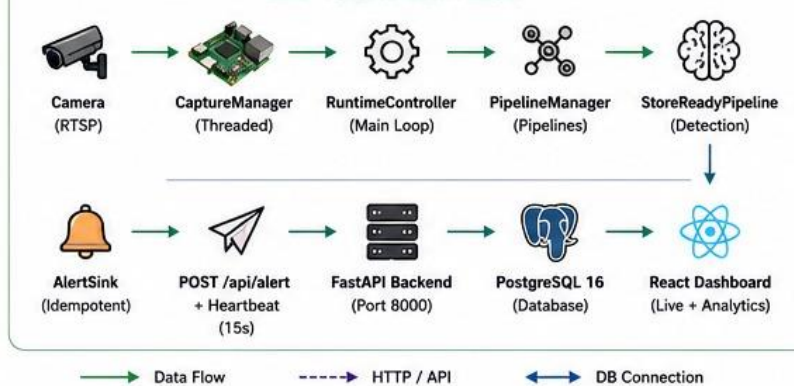


## AI MODELS & USE CASES MAPPING

| Model                              | Purpose                               | Use Cases (UC)                                                                                                                |
|------------------------------------|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| FM1.onnx (Person Detection)        | Detect all persons                    | UC-01 Crowd, UC-03 Intruder, UC-07 Footfall, UC-08 Dwell, UC-09 Engagement, UC-11 Most Visited, UC-12 Staff Mobile + Customer |
| SM3.onnx (Staff Detection)         | Detect staff / salespersons           | UC-01, UC-05 Staff on Mobile, UC-06 SP Gathering, UC-08, UC-09 Engagement, UC-10 Desk Presence, UC-12 Staff Mobile + Customer |
| YOLO11x.onnx (Phone Detection)     | Detect mobile phones (COCO 67)        | UC-05 Staff on Mobile, UC-12 Staff Mobile + Customer                                                                          |
| YOLO11n.onnx (Person Detection)    | Fast person detection (Gate footfall) | UC-07 Footfall IN/OUT                                                                                                         |
| prop.onnx (NOT USED) (Wrong Model) | Gun detector (1 class, incorrect)     | NOT USED<br>UC-04 uses Classical CV (SSIM + abs_diff)                                                                         |

UC-04 Prop Monitoring uses Classical CV against calibrated baseline images. Models used only where appropriate (people, staff, phone).

## END-TO-END DATA FLOW



## SECURITY ARCHITECTURE



## LEGEND



## KEY FEATURES

- Modular runtime (Capture → Pipeline → Alert)
- Schedule + interval based execution
- Idempotent alerts (SHA-256 event\_id)
- 15s heartbeat for device health
- Scalable to 54+ cameras
- Dual authentication (Cookie + API Key)
- Classical CV + ONNX hybrid
- Production ready & extensible

## NOTES

- All Raspberry Pis run main\_combine\_detection.py (v5.8)
- Each Pi has unique DEVICE\_ID (see pi\_config.py)
- System service: sen.co\_detection.service (Restart=always)

## TOTAL SUMMARY

Stores: 6  
Raspberry Pi Cameras: 54  
Use Cases (UC): 12  
AI Models: 5  
Status: Phase 1 Live (C006 Moulali)



# THANK YOU

Transforming Vision into Intelligence  
**ELVA by ESTPL**

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